

Skill - 11

1. Command History

```
harshika@siri: ~  
harshika@siri:~$ gedit skill11.c  
** (gedit:3944): WARNING **: 03:24:26.701: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
** (gedit:3944): WARNING **: 03:24:26.701: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found  
harshika@siri:~$ gcc skill11.c -o skill11  
harshika@siri:~$ ./skill11  
Simple Shell - Skill 11  
myshell> ls  
Command stored: ls  
myshell> pwd  
Command stored: pwd  
myshell> data  
Command stored: data  
myshell> history  
Command History:  
1 ls  
2 pwd  
3 data  
4 history  
myshell> one  
Command stored: one  
myshell> two  
Command stored: two  
myshell> three  
Command stored: three  
myshell> history  
Command History:  
1 history  
2 one  
3 two  
4 three  
5 history  
myshell> ls | wc  
Pipeline layout:  
Command 1: ls  
Command 2: wc
```

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3 #include <string.h>  
4 #include <unistd.h>  
5 #include <sys/wait.h>  
6  
7 #define HISTORY_SIZE 5  
8 #define INPUT_SIZE 200  
9 #define MAX_COMMANDS 10  
10  
11 char history[HISTORY_SIZE][INPUT_SIZE];  
12 int history_count = 0;  
13  
14 /* Store command in history */  
15 void add_history(char *command)  
16 {  
17     if (strlen(command) == 0)  
18         return;  
19  
20     if (history_count < HISTORY_SIZE)  
21     {  
22         strcpy(history[history_count], command);  
23         history_count++;  
24     }  
25     else  
26     {  
27         /* Shift old commands */  
28         for (int i = 1; i < HISTORY_SIZE; i++)  
29         {  
30             strcpy(history[i - 1], history[i]);  
31         }  
32         strcpy(history[HISTORY_SIZE - 1], command);  
33     }  
34 }  
35  
36  
37 /* Display command history */  
38 void show_history()  
39 {  
40     if (history_count == 0)  
41     {  
42         printf("History is empty.\n");  
43         return;  
44     }  
45  
46     printf("\nCommand History:\n");  
47  
48     for (int i = 0; i < history_count; i++)  
49     {  
50         printf("%d %s\n", i + 1, history[i]);  
51     }  
52 }
```